ÖNEMLİ

① $\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{x}{\sin x} = \lim_{x\to 0} \frac{\tan x}{x} = \lim_{x\to 0} \frac{x}{\tan x} = 1$

② $\lim_{x\to 0} \frac{\sin ax}{bx} = \lim_{x\to 0} \frac{ax}{\sin bx} = \lim_{x\to 0} \frac{\tan ax}{bx} = \lim_{x\to 0} \frac{ax}{\tan bx} = \frac{a}{b}$ *

* $\lim_{x\to 0} \frac{\sin ax}{\sin bx} = \lim_{x\to 0} \frac{\tan ax}{\tan bx} = \lim_{x\to 0} \frac{\sin ax}{\tan bx} = \frac{a}{b}$

③ $\lim_{x\to a} f(x) = 0 \overset{=}{} \Rightarrow \lim_{x\to a} \frac{\sin f(x)}{f(x)} = \lim_{x\to a} \frac{\tan f(x)}{f(x)} = 1$ $\overset{=}{}$

(numerator $\to 0$, denominator $\to 0$)

* $\frac{\sin \square}{\square} \to 1$ $\qquad \frac{\tan \square}{\square} \to 1$

④ $\lim_{x\to 0} \frac{1-\cos x}{x} = \lim_{x\to 0} \frac{\cos x - 1}{x} = 0$ $\qquad \frac{1-\cos \square}{\square} \to 0$

⑤ $\lim_{x\to \infty} \frac{\sin x}{x} = 0$ $\qquad \frac{\sin \infty}{\infty} \to 0$

⑥ $\lim_{x\to 0} x \cdot \sin\frac{1}{x} = 0$ $\qquad 0 \cdot \sin\infty \to 0$

⑦ $\lim_{x\to \infty} x \cdot \sin\frac{1}{x} = \lim_{x\to \infty} \frac{\sin\frac{1}{x}}{\frac{1}{x}} = 1$

$\downarrow$

$\frac{\sin \square}{\square} \to 1$

⑧ Üstel ve Logaritmik Limitler

★ $\lim_{x \to \infty} \ln x = \infty$ $\quad$ $\lim_{x \to 0^+} \ln x = -\infty$

★ $\lim_{x \to \infty} e^x = \infty$ $\quad$ $\lim_{x \to -\infty} e^x = 0$

★ $0 < a < 1$ için

$\lim_{x \to \infty} a^x = 0$ $\quad$ $\lim_{x \to -\infty} a^x = \infty$

★ $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$

★ $\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1$

$\frac{e^{\square} - 1}{\square} \to 1$ $\quad$ $\frac{\ln(1+\square)}{\square} \to 1$

⑨ Ters Trig. Fonk. Limitleri

$\lim_{x \to \infty} \text{Arctan}\, x = \frac{\pi}{2}$ $\quad$ $\lim_{x \to -\infty} \text{Arctan}\, x = -\frac{\pi}{2}$

$\lim_{x \to 1^-} \text{Arcsin}\, x = \frac{\pi}{2}$ $\quad$ $\lim_{x \to -1^+} \text{Arcsin}\, x = -\frac{\pi}{2}$

$\lim_{x \to 1^-} \text{Arccos}\, x = \frac{\pi}{2}$ $\quad$ $\lim_{x \to \infty} \text{Arccosec}\, x = 0$

Türev Tablosu

1. $y = x^n \rightarrow y' = nx^{n-1}$
2. $y = \cos x \rightarrow y' = -\sin x$
3. $y = \sin x \rightarrow y' = \cos x$
4. $y = \tan x \rightarrow y' = 1 + \tan^2 x = \sec^2 x$
5. $y = \cot x \rightarrow y' = -(1 + \cot^2 x) = -\operatorname{cosec}^2 x$
6. $y = \sec x \rightarrow y' = \sec x \cdot \tan x$
7. $y = \operatorname{cosec} x \rightarrow y' = -\operatorname{cosec} x \cdot \cot x$
8. $y = \ln x \rightarrow y' = \frac{1}{x}$
9. $y = e^x \rightarrow y' = e^x$
10. $y = a^x \rightarrow y' = a^x \cdot \ln a$
11. $y = \operatorname{Arcsin} x \rightarrow y' = \frac{1}{\sqrt{1-x^2}}$
12. $y = \operatorname{Arccos} x \rightarrow y' = -\frac{1}{\sqrt{1-x^2}}$
13. $y = \operatorname{Arctan} x \rightarrow y' = \frac{1}{1+x^2}$
14. $y = \operatorname{Arccot} x \rightarrow y' = -\frac{1}{1+x^2}$

İntegral Tablosu

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
2. $\int \cos x \, dx = \sin x + C$
3. $\int \sin x \, dx = -\cos x + C$
4. $\int \sec^2 x \, dx = \tan x + C$
5. $\int \operatorname{cosec}^2 x \, dx = -\cot x + C$
6. $\int \tan x \, dx = -\ln|\cos x| + C$
7. $\int \cot x \, dx = \ln|\sin x| + C$
8. $\int \sec x \, dx = \ln|\sec x + \tan x| + C$
9. $\int \operatorname{cosec} x \, dx = -\ln|\operatorname{cosec} x + \cot x| + C$
10. $\int \frac{dx}{x} = \ln|x| + C$
11. $\int e^x dx = e^x + C$
12. $\int a^x dx = \frac{a^x}{\ln a} + C$
13. $\int \cos^2 x \, dx = \int \frac{1+\cos 2x}{2} dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$
14. $\int \sin^2 x \, dx = \int \frac{1-\cos 2x}{2} dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$